Python-Functions

Objective

- Global and local variables
- Map Function
- Filter Function
- Other Built-In Functions

Global and local variables

In python, the variables are defined with the two types of scopes.

1. Global variables
2. Local variables

The variable defined **outside** any **function** is known to have a **global scope**, whereas the variable defined **inside a function** is known to have a **local scope**.

Example1:

```
def print_message():
    # the variable message is local to the function itself
    message = "hello !! "
    print(message)

print_message()

print(message) # this will cause an error since a local variable cannot be accessible here.
```

Example 2:

```
# Global Variable
def calculate(*args):
    sum=0
    for arg in args:
        sum = sum +arg
    print("The sum is",sum)

sum=0

calculate(10,20,30) #60 will be printed as the sum
print("Value of sum outside the function:",sum) # 0 will be printed
```

Output:

The sum is 60

Value of sum outside the function: 0

Note:

you can not access local variable outside function and even global variable inside function(if you have variable present in function with same name). to use gloabal variable inside function you have to use global keyword.

```
x = "global "
def foo():
    global x
    y = "local"
    x = x * 2
    print(x)
    print(y)

foo()
```

Output:

```
global global
local
```

Map Function

The map() function executes a specified function for each item in an iterable. The item is sent to the function as a parameter.

Syntax:

map(function, iterables)

function :- Required. The function to execute for each item

iterable :- Required. A sequence, collection or an iterator object. You can send as many iterables as you like, just make sure the function has one parameter for each iterable.

Example:

```
def myfunc(n):
  return len(n)

x = map(myfunc, ('apple', 'banana', 'cherry'))
print(x)

#convert the map into a list, for readability:
print(list(x))
```

Output:

```
<map object at 0x0214C880>
[5, 6, 6]
```

Example 2:

```
# Python program to demonstrate working of map.
# Return double of n
def addition(n):
    return n + n

# We double all numbers using map()
numbers = (1, 2, 3, 4)
result = map(addition, numbers)
print(list(result))
```

Output :

[2, 4, 6, 8]

Filter Function

The filter() function returns an iterator were the items are filtered through a function to test if the item is accepted or not.

Syntax

filter(function, sequence)

Parameters:

function: function that tests if each element of a sequence true or not.

sequence: sequence which needs to be filtered, it can be sets, lists, tuples, or containers of any iterators.

Returns:

returns an iterator that is already filtered.

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Example

Filter the array, and return a new array with only the values equal to or above 18:

```
ages = [5, 12, 17, 18, 24, 32, 3]
def myFunc(x):
 if x < 18:
  return False
 else:
  return True

adults = filter(myFunc, ages)
for x in adults:
 print(x)
```

```
Output:

18
24
32
```


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Other Built-In Functions

abs()	dict()	help()	min()	setattr()
all()	dir()	hex()	next()	slice()
any()	divmod()	id()	object()	sorted()
ascii()	enumerate()	input()	oct()	staticmethod()
bin()	eval()	int()	open()	str()
bool()	exec()	isinstance()	ord()	sum()
bytearray()	filter()	issubclass()	pow()	super()
bytes()	float()	iter()	print()	tuple()
callable()	format()	len()	property()	type()
chr()	frozenset()	list()	range()	vars()
classmethod()	getattr()	locals()	repr()	zip()
compile()	globals()	map()	reversed()	__import__()
complex()	hasattr()	max()	round()	
delattr()	hash()	memoryview()	set()	

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Summary

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Thank You